

Improving Mental Health Among Working-Class Indian Women: Insight From An Interview Study

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ABSTRACT

Mental health applications (apps) show a good potential to help people suffering from a variety of mental illnesses. Most existing apps focus on people from developed countries. Our research contributes by focusing on underserved populations - Indian Working-Class Women. This study aims at creating a stress and anxiety management application for working-class Indian women (people who are a part of the workforce). We conducted a one-on-one interview with 31 working-class Indian women. Participants highlighted household chores and workload as major sources of stress/anxiety for them. The results also highlight several features required in mental health apps by this group including mood tracking features, social community features. Participants highlighted some concerns with existing mental health apps including privacy issues and the high fees charged for premium features. Based on our findings, we discuss design implications for future work in the field of creating interactive mHealth apps.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction design; Interaction design process and methods; User centered design.

KEYWORDS

mental health, user-centered design, working-class Indian women, HCI, second-shift phenomenon

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1 INTRODUCTION

In 2011 the American Psychological Association warned that stress is becoming a public health crisis [1]. Stress is a growing concern, and everybody requires a different treatment for stress as every individual is unique. Attending clinical sessions is not feasible for

most people in developing and low-resource countries. Also, some individuals are not comfortable sharing their thoughts with any other individual, making it more challenging to reach the root cause of stress. In a developing country like India discussing topics such as mental health is still considered taboo [17]. Thus, people going through mental distress ignore sharing it with their family members or life partners so as not to become the cause of their stress and cause panic [13]. In such communities, women are more vulnerable to mental health problems than men because of the many responsibilities they play in society, including work, childcare, housewife, and more - gender role [10]. Specifically, the second shift phenomenon (i.e., the housekeeping and childcare responsibilities that follow a workday outside the home) [6, 11] is a major cause of stress and anxiety for working-class women, as a women's life never has an "off" day. Furthermore, women lose a large chunk of sleeping hours because they work with a busy schedule [16], trying to meet up with both work and home caring requirements which are usually not considered as work.

Stress can lead to more severe issues, such as depression, if not addressed at an early stage [15]. Thus, getting help at an early stage is essential to treating stress. In addition, preserving individuals' identity is a key concern when individuals share their mental health issues, hence, most people in developing countries are skeptical to seek help even from a health professional due to the stigma associated with it. Mental health apps are helpful to manage stress given the pervasiveness of mobile phones. Mental health apps have a huge potential to help people suffering from a variety of mental illnesses when they need them. They are also affordable compared to conventional mental health treatments. Nonetheless, there is a paucity of studies that focuses on mHealth interventions for stress among women. More specifically, while most existing apps focus on designing for people from developed countries, our research contributes by filling that gap via focusing on underserved populations - Indian Working-Class Women (people who are part of the workforce). According to Almeida et al. [2], there are very few user-centric design approaches that aim solely at women's mental health. One of the challenges is figuring out how to create interventions that are both helpful and acceptable to women who are suffering from mental health issues, especially those from developing countries where there is still a lot of stigmas associated with mental health and gender role is still a major issue.

We aim to investigate how the design of mobile apps can support working-class Indian women in managing their mental health by investigating their mental health struggles, factors associated with mental health, strategies they use to manage their mental health, and their requirements for apps that would help them with their mental health. This paper reports on the results from the first

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study that aims to apply the user-centric design approach [9] to the design and development of an app for managing stress and anxiety in working-class Indian women. To achieve our goal, the study is divided into four stages: First, we advertised the study over LinkedIn, various female groups on Facebook, and over WhatsApp. Second, we disseminated a questionnaire to collect demographic information and individuals' awareness of mobile apps for mental health and sent it to those women who showed interest. Third, based on the results of this questionnaire, we invited participants who met the inclusion criteria for one-on-one interviews, following our ethics approval. We asked questions about their mental health issues, events that trigger mental health, factors that contribute to mental health issues, strategies used in managing their mental health issues. Finally, we asked their opinion about two popular mental health apps which they were required to use before the interview (Headspace [18] and Happify [19]); the features and functionalities they liked, and the features and functionalities they disliked. We interviewed 31 working-class Indian women. The interviews were recorded and transcribed with the permission of the participants. Then, we conducted a thematic analysis of the collected data. Our results revealed that participants used the following strategies to deal with their mental health issues: doing something to distract themselves from their current negative mood, using relaxation exercises and methods to relieve symptoms, trying to be alone. They also highlighted household chores and workload as major sources of stress/anxiety for them. The results from our analysis highlighted several features required in an app to promote mental health including mood tracking features, social community, or social support features. Participants highlighted some concerns with existing mental health apps including privacy issues and the high fees charged for premium features. Based on our findings, we discuss design implications for future work in the field of creating interactive mHealth apps, with a focus on mental health interventions targeting women in developing countries.

2 RELATED WORKS

Human-Computer Interaction (HCI) research on mental health has gained increasing attention. mHealth technology has been found as a potential solution for assisting users with their health care needs. Researchers have increasingly started to employ mobile phones as digital solutions for delivering health treatments in recent years [8]. Research has found that considering the target users in the design of the mHealth apps is critical to understand, accommodate, and design apps that meet their needs [3, 9]. This section reviews recent work in the domain of mHealth apps designed for addressing mental health issues.

Kenny et al. [7] conducted a study using five focus groups to explore adolescents' needs and concerns with mental health mobile apps. The participants identified "safety, engagement, functionality, social interaction, awareness, accessibility, gender, and giving young people control" as important factors to consider in designs. Similarly, Alqahtani et al. [3] held six focus group sessions with participants who self-diagnosed as having mental health problems. The results show that essential aspects that users seek in a mental health app are basic design features such as "personalization of the app, including tracking and feedback, live support, and social

community; and providing motivational content and relaxation exercises". Cheng et al. [5] investigated the best approach to arrange and show a psychoeducational module delivery app. The authors conducted a participatory design workshop and user experience interviews with 40 participants and 15 participants, respectively. The results revealed that participants prefer activities that required active participation and provided opportunities to practice valuable skills. Participants also raised concerns about the presentation of material and wanted to personalize their app experience. Peter et al. [14] undertook participatory research to learn about workers' opinions, preferences, and ideas to create a mental health app that is both interesting and successful in enhancing emotional well-being in male-dominated companies. They found that accessible languages, simplicity of use, visual appeal, and offline mood were all important aspects for mental health apps that users wanted. There is observed evidence suggesting that women suffer much more from mental health issues than men, even after adjusting for perceived work circumstances, and gender roles, which is not true for physical symptoms [12]. Even though a significant amount of research has been done to understand users' impressions of mental health app design, the above-mentioned gender difference has not been considered broadly and more importantly among women in developing countries where significant gender differences exist. This study primarily focuses on understanding the strategies used by Indian female participants (who have experienced/experiencing mental health issues) to deal with mental health issues in their lives, their perspectives, and opinions about popular mental health apps, and how these strategies and ideas could be leveraged in the design of a mental health app, all of which have received little attention in previous studies. Our research offers some insight into addressing this limitation as described in the next section.

3 METHOD

Upon receiving clearance from our institutional ethics review board, we advertised the study over LinkedIn, various female groups on Facebook, and over WhatsApp. Participants completed a pre-study questionnaire and one-on-one interviews. We disseminated a questionnaire to get demographic information and individuals' awareness of mHealth apps for mental health from the participants that showed interest. Based on the results of this questionnaire, we selected participants who met the inclusion criteria, and invited them for the interviews. To analyze the interview data, we performed a thematic analysis [22]. This section discusses the procedures we followed to collect and analyze our data.

3.1 Selection Criteria

We conducted one-on-one interviews with 31 working-class Indian women who were experiencing or have experienced mental health issues like stress or anxiety or depression, or mood swings based on self-diagnosis. The inclusion criteria for this study were:

- Working-class Indian women living in India; knowing basic English and of age 18 years and older
- Those who are experiencing or have experienced mental health issues such as stress or anxiety or low mood or negative feelings or thoughts, worry, fear, panic attack, or depression based on self-diagnosis.

3.2 Participants

37 participants completed the questionnaire, among which 31 met the inclusion criteria. Out of these 31 participants, 73% mentioned that they have never used any mental health app to support themselves, and only 27% used such apps. Participants ages were between 18 and 54 years old, distributed as follows: 18-24 (3.4%), 25-34 (74.1%), 35-44 (19.3%), and 45-54 (3.2%). The majority of the participants (81%) self-identified as married, while 3% were widowed and 16% were single. 81% of the participants hold a Master's degree, and 19% hold a bachelor's degree. All the participants were employed.

3.3 Interview Procedure

After selecting participants who met the inclusion criteria, we invited them for a 30-minute interview. All interviews were conducted over Microsoft Teams, which is approved by the research ethics board of our university. As mentioned above, the majority of participants had never used a mental health app. So, we asked all participants to explore two mental health apps (Headspace [18] and Happify [19]), before the interview. Using these two apps familiarized them with the study context. We selected these two apps because they are popular with 10M+ and 500K+ downloads and with a rating of 4.3 over Play Store and App Store for Headspace and Happify, respectively. Also, they have been vetted by mental health specialists and are available on the website of the Anxiety and Depression Association of America [20]. The interview questions were divided into two phases. In the first phase, we asked the participants questions such as whether they have or had any mental health issues, the major causes, how they tackle stress and what are their views on how Indian society handles mental health. In the second phase of the interview, participants were asked about their views on the Headspace and Happify app which they explored before the interview, design suggestions, the features they would want to see in the app for managing their stress and anxiety, and their motivation to use a mental health app. Various themes emerged from these two interview phases, and they are presented in Section 4.

3.4 Data collection and Analysis

Interviews were voice recorded and transcribed using Microsoft teams, and all the transcribed data were corrected (grammatically and spelling-wise) using the recordings. Then all the corrected answers were fed into NVivo 12 Pro, a qualitative and mixed-methods research software tool used to analyze the unstructured text, audio, video, and picture data, such as interviews, focus groups, and surveys [21]. Data were scattered into different sections, Nvivo12 further aided the thematic analysis. We used an inductive approach [4] to generate the codes. A thematic analysis was chosen because it allows us to evaluate a big data set methodically, revealing patterns in the text while considering the context of what participants said, allowing us to interpret the data more precisely.

4 RESULTS

In this section, we present the results of the themes generated from the interviews as qualitative data.

4.1 Phase1

In this phase, our thematic analysis uncovered three major themes: events that trigger stress/anxiety among participants, strategies, and activities used to manage stress/anxiety, and various social and personal factors that play a major role in participants' mental health. The following subsections present each one of these themes and provide sample quotes as examples.

4.1.1 Things that trigger stress/anxiety. This theme reflects the things that trigger stress/anxiety among the participants. Based on our analysis, all participants highlighted household chores, workload, taking care of kids, and when things go unplanned as a source of stress/ anxiety. For example,

"In my day-to-day life, I think I get the most stressed **when my kids have exams** or when you know there is some deadline or if they have some projects as now everything revolves around the kids..." [P5].

"I'm having a hectic schedule, even if I work from home. Still, I'm not able to have a proper **work-life balance**." [P11].

"A very stressful thing that I have noticed is like when something is **not going according to the plan**." [P13].

"I am deficient in managing pressure I **cannot work under pressure**. OK, so given my current situation like currently what I'm going through. I do have some things that are causing anxiety and not being able to function as I should." [P17].

4.1.2 Habits/Activities to manage stress/anxiety. This theme reflects the habits/activities that help the participants to calm themselves. Based on our analysis, few participants try to ignore the mental health issues altogether, few watch entertainment, and others prefer staying alone. For example,

"So, when I am in such a state of mind, I **prefer to be alone** I sort of lock myself up in a room and **listen to some music** or read something. which cheers me up." [P6].

"I don't seek help very easily, so I talk to myself I cool myself down, **I watch stand-up** shows so that I can pull myself and come back to normal. That's what I do." [P7].

"Uh, I usually **ignore the things** which are bothering me..." [P30].

4.1.3 Social and personal factors that play major roles in mental health. This theme reflects the various social and personal factors that play a major role in our participants' mental health. Based on our analysis, support from family/the life partner, salary and work appraisals, and the view of Indian society about mental health play a major role in people's mental health. For example,

"I think one **feels at ease** when you see you are credited with some amount on a particular date. It makes you happy whenever you get some appraisals." [P2].

"I say if you have a **supporting partner** there's nothing like it. I mean, if you have somebody on whom you can rely to help you with your chores be it a small thing. It makes a lot of difference in a woman's life." [P6].

"I think, most of the women are not getting a job that they are expecting right, and also **many families are not** supporting women." [P9].

"...it's very easy for **people to judge** and they just say, it's because if she is unable to do anything let her stick to the boundaries of household.." [P12].

4.2 Phase2

In this phase, our thematic analysis resulted in defining three themes: people's views about popular existing mental health apps - Headspace and Happify, views about features of these apps, and their motivation to use mental health apps. The following subsections define each one of these themes and provide sample quotes as examples.

4.2.1 Things that trigger stress/anxiety. As mentioned above, 73% of participants have never used a mental health app to support themselves. Therefore, making them go through these two apps gave them an idea of what to expect from a mental health app. This theme reflects what their first impression of these two apps was. In summary, participants find the apps appealing, easy to use, they also liked that the apps provided relaxation exercises and that the apps revealed and provided opportunities for the participants to reflect on things that may be a source of stress to them. However, they also complained and felt disappointed that most of the important features in the app require premium subscriptions for a fee. They also highlighted the need for guidance on how to do meditation, which is essential for beginners. For example,

"So, about the headspace app, when I first opened it, I think it has a wonderful UI. That's what I want, it's **very appealing**. It **feels very realistic**, also it was **very easy** to log in. However, most of the things required the premium..." [P3].

"Yeah, so you have that questionnaire in these apps which make you think what are the different stress points in your life. So, when I was reading those questions it **revealed a lot about me**, even those things that I did not know which could be the reasons of stress." [P6].

"I did choose happify. I've tried it. Yes, I like the breath in and breathe out part. So that is good. **I thought it was** relaxing for me." [P9].

However, few participants were disappointed as they did not expect mental health apps to charge them for managing their stress/anxiety. For example,

"I installed headspace app a long back ago but **I wasn't able to cope with it**. The reason was that again and again, it was asking for payment; secondly, I wasn't able to meditate because this topic is not that easy. So, having lengthy meditations is very tough for a beginner, so for me also it didn't work like that." [P13].

4.2.2 Views on features for the app. This theme reflects participants' views of the features for the two mental health apps based on their requirements and expectations. Participants appreciated mostly, the Mood tracker and the Social community features of the apps. For example,

Mood tracker: "Mood tracker would be something that will help me to know myself, how am I doing and how do I react to things. So, I think if something can keep track of that, I'll know myself better..." [P2]

Social Community: "...I might be always active in the forums and you know, like when women work together, they do wonders. I just love that..." [P1].

Participants also found that voice memos are more attractive than writing. For example,

"**Voice memos** are attractive. Writing a diary or a journal, some people do that religiously, but most people don't write these days, so voice memo thing might be a good feature for today's generation." [P6].

"Nowadays you don't have many people with whom you can speak or sometimes you just shy away because you don't want to share your problems with the people you know. It is good to know that you can **just vent out your emotions** or your feelings, even when nobody is listening..." [P8].

Finally, privacy is a primary concern for most users, especially when it comes to mental health issues. The data collected shows that participants have concerns about sharing their information. For example,

"I think **I won't share my details here**; I think I will go for hiding my identity because these days you know how much our privacy is getting invaded." [P15].

4.2.3 Motivation to use the mental health app. This theme reflects why people think they would be motivated to use the mental health app. In general, people are motivated to use the app to help them with various mental health issues. The following are some quotes from the interviews showing participants' negative experiences for using the app; explaining why participants were motivated to use the app:

"**You know having a bad day**, but I do not necessarily need to have a bad day to open this app. It could be anything. It could be something you know, recreational that I want to do just another past time activity that makes me feel happy by the end of the day" [P1].

"Whenever I will **feel stressed** or a bit low or unhappy, I will automatically go to the app and will just start using it..." [P4].

5 DISCUSSION

The goal of this research was to gain a better understanding about the viewpoints, preferences, opinions, requirements, and strategies that working-class Indian women use to deal with mental health concerns such as stress, depression, or anxiety. Given the well-documented challenges of engaging individuals in mental health promotion, insights from this study are critical for designing and developing successful mental health intervention apps. Our results highlight many interesting and important insights including that most participants were unaware of the availability of mHealth apps to support their mental health. In this section, we discuss our findings, specifically, participants' experience with mental health, strategies used in coping with mental health issues, and expectations from mHealth apps.

5.1 Participants experience mental health issues

All the participants agreed that they had gone or are going through stress/anxiety. It was mostly due to their workload and less time to devote to themselves. It was brought up, many times, that a big

chunk of their time is dedicated to their work and family that they forgot what they enjoyed. For example, one participant explained that *“...I used to play badminton and it has been 10 years since I played it. Last month my son asked me, whether I know how to play badminton since I was doing so well that he was shocked to see me playing such a good game. I forgot that I know how to play that game that efficiently...”* [P5]. These findings indicate that not being able to dedicate time for self-care can become a root cause of losing control over mental health. If we talk about the reasons that trigger stress as discussed in section 4.4.1, participants mentioned reasons like when their kids have exams or when there is a lot of work pressure. Mental health app designers should consider adding a reminder strategy more frequently for users to take care of themselves based on the activities that are suitable for them or the app can also assist participants in planning the activities in a way that leaves them some time for self-care. To personalize mental health apps, designers can suggest strategies using participants’ profiles of what works for them. They can also recommend new strategies that have worked for others in a similar way and evaluate the effectiveness for that particular individual and if effective, include them in the list of effective strategies in the profile. With respect to strategies used in dealing with mental health issues, as mentioned in the result section, participants had different ways of handling stress, so it is not wise to use a one-size-fits-all approach. Hence, designers of mental health apps should take into consideration the way their audience deals with mental health issues as this can help the users to achieve the desired results. The three popular strategies used by our participants to handle stress include: (1) doing something to distract themselves from their current negative mood, (2) using relaxation exercises and methods to relieve symptoms, and (3) trying to be alone. Designers of mental health apps targeting this group of users should include these strategies as part of their app features.

5.2 Participants’ experience with mental health issues

Participants with mental health concerns typically seek assistance from their social circle, especially those with similar experiences who will not judge or stereotype them. As a result, the inclusion of a social community in the investigated app matches the users’ method of coping with their challenges in real life. However, most participants do not want to connect the app to any social media platform or disclose their identity in the social community within the app. Nevertheless, participants want the app to provide a mechanism to chat anonymously. Anonymity will give them a sense of security, and everyone will be able to support each other more actively within the app. Therefore, we suggest that mental health app designers should consider these needs and allow users to be part of social communities with the benefit of protecting their identity as necessary so that people with mental health issues do not feel overwhelmed. According to the participants having a mood tracker will allow them to keep track of any changes in their mood, reveal likely triggers of mental issues, and help to keep their overall mood on the brighter side (mood management). Moreover, participants were also positive about keeping a diary/voice memo as they thought that it will help them to vent out all the negativity and nobody will know. This shows that the tracking feature is very important for

mental health apps targeting this group, as it helps them keep track of their behavior and mood patterns. Designers can think of various innovative ways to implement trackers to reduce the boredom and tediousness often associated with them. One way suggested by the participants is the use of voice-based tracking in the form of voice memos as opposed to writing.

Privacy is a primary concern for most users, especially when it comes to mental health issues. Therefore, designers must adhere to high privacy standards and make their data handling and privacy policy transparent to earn the trust and increase the likelihood that their app will be accepted by the target users.

When asked about the motivation to use the mental health app, participants mentioned a variety of ways they would like to use the app. Some individuals prefer to use them just when they are in a dilemma, whereas others said they would prefer to use the app regularly. Therefore, the app should be designed to support various types of usage scenarios and user groups.

6 FUTURE WORK

Based on the rich data collected through these one-on-one interviews, we are planning to create a stress and anxiety management app incorporating our findings concerning the app features. Then, we plan to evaluate the effectiveness of the app with our target audience (working-class Indian women).

7 CONCLUSION

This study is part of a larger effort aimed at developing and evaluating the effectiveness of a mobile app for mental health promotion. The findings shed light on Indian women’s perspectives, beliefs, and preferences for mobile apps to assist them with mental health and how such apps should be created. It also sheds light on various factors contributing to mental health issues among our target audience and the strategies they used to deal with the issues. According to our findings, participants utilized various strategies to deal with their mental health difficulties like staying alone, relaxation exercises, and ignoring the things that overwhelm them via using techniques such as distraction. Understanding end users’ requirements and concerns regarding mental health apps can help designers and developers create mental health apps that are useable, valuable, acceptable, and effectively utilized by the target audience to enhance mental health and emotional well-being in the future. Most existing apps focus on people from developed countries, our research contributes by filling that gap via focusing on underserved populations - Indian Working-Class Women.

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